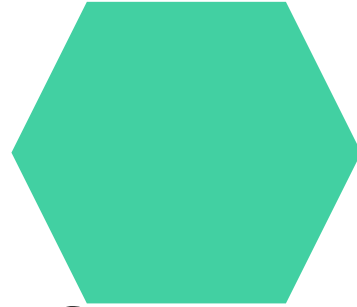
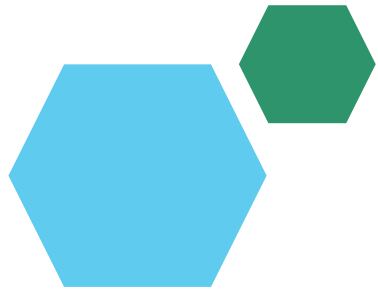




Project Title - Seasonal Agriculture Performance Analysis

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College Name- St. Vincent Pallotti College of Engineering and Technology Nagpur

AICTE STU ID : STU650311bdd263a1694699965



PROBLEM STATEMENT

- Agricultural operations in India are deeply intertwined with seasonal environmental changes, input resource availability, and varying market conditions.
- Raw farm data masks critical dynamics, obscuring how weather, irrigation methods, and soil chemistry interact to impact farm solvency and output.
- Significant unpredictability exists across seasons—especially between the monsoon-fed Kharif cycle and the arid summer Zaid cycle.
- **Core Objective:** Build an end-to-end data processing and visualization pipeline to isolate the primary agronomic drivers of yield, quantify irrigation efficiencies, and expose seasonal economic vulnerabilities for evidence-based policymaking.



Project Description

- **Analytical Scope:**

- Analyzed a multi-regional agricultural dataset of **4,000 farm records** containing 28 environmental, soil, agronomic, and financial parameters across Kharif, Rabi, and Zaid seasons.

- **Key Methodology:**

- **Data Wrangling:** Handled missing values in Rainfall_mm, Soil_Moisture_pct, and Yield_Tonnes_Ha via crop-and-season-grouped median imputation to prevent distributional skew.
- **Feature Engineering:** Formulated critical domain metrics including **Profit Margin Percentage**, **Cost per Hectare**, and **Water Productivity Ratio** (Tonnes produced per 1,000 m³ applied).
- **Exploratory Data Analysis:** Conducted statistical evaluations using correlation matrices, multi-distribution violin curves, non-linear regression curves, and 4D bubble charts.

WHO ARE THE END USERS?

- **Farmers & Agronomists:** Gain data-backed advisory on optimal crop-season pairings, input timing, and water requirements to avoid negative off-season margins.
- **Irrigation & Water Resource Authorities:** Access empirical benchmarks on water loss under flood systems versus micro-irrigation to allocate seasonal water quotas effectively.
- **Agricultural Extension Officers & Agritech Startups:** Deploy localized predictive insights for pest outbreak management and precision resource planning.
- **Policy Makers & Credit/Insurance Institutions:** Establish objective risk-tiering for crop insurance premiums, minimum support prices (MSP), and targeted seasonal fertilizer subsidies.

Technology Used

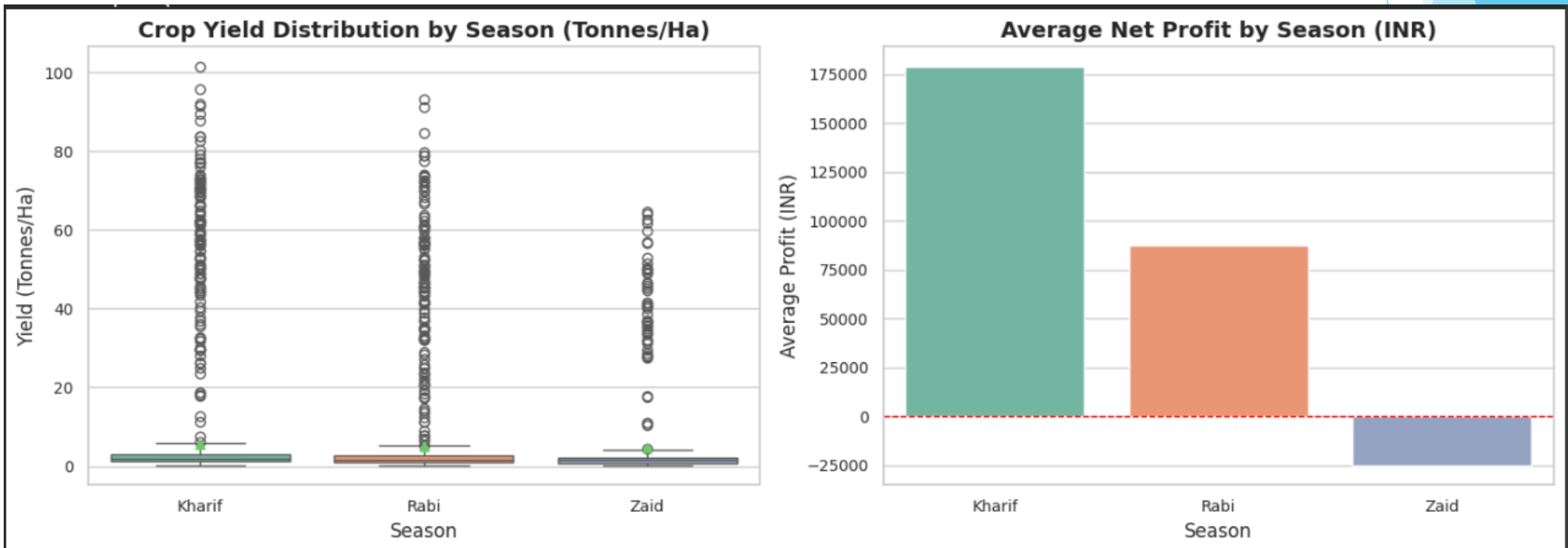
- **Language & Runtime:** Python 3.x executed on Google Colaboratory (cloud-hosted Jupyter environment).
- **Data Manipulation & Computation:**
 - pandas for multi-condition querying, grouped transformations, and missing-value imputation.
 - numpy for vectorized mathematical operations and efficiency metric calculations.
- **Data Visualization & Analytics:**
 - matplotlib for custom chart axes, grid styling, and publication layouts.
 - seaborn for statistical visualizations (violin density plots, box plots, and correlation heatmaps).
- **Version Control & Documentation:** Git and GitHub for codebase versioning and reproducibility.



RESULTS

Key Takeaways:

- **Kharif Dominance:** Kharif crops achieved the highest average yield (**5.64 tonnes/ha**), representing a **20.8% yield advantage** over summer Zaid cultivation.
- **Seasonal Environmental Support:** Kharif's high yield is strongly supported by abundant average precipitation (852.1 mm) and sustained atmospheric humidity.
- **Rabi Stability:** Winter Rabi crops demonstrated consistent yields (**5.08 tonnes/ha**) with lower standard deviation across staple cereals (Wheat and Maize).
- **Zaid Constraints:** Summer crops averaged the lowest yield (**4.67 tonnes/ha**) due to high evapotranspiration rates and soil moisture deficits.



RESULTS

Key Takeaways:
Drip Irrigation Outperforms: Drip systems achieved the highest water productivity (**5.88 tonnes per 1,000 m³** of water used).
Inefficiency of Flood Systems: Traditional flood irrigation consumed over 4,100 m³ on average while generating a water efficiency score of only **3.95**, demonstrating heavy resource loss.
Pest & Disease Risk Correlation: High soil moisture combined with poor drainage under flood irrigation led to an elevated disease/pest risk (**46.8%**) compared to micro-irrigation (**28.4%**).

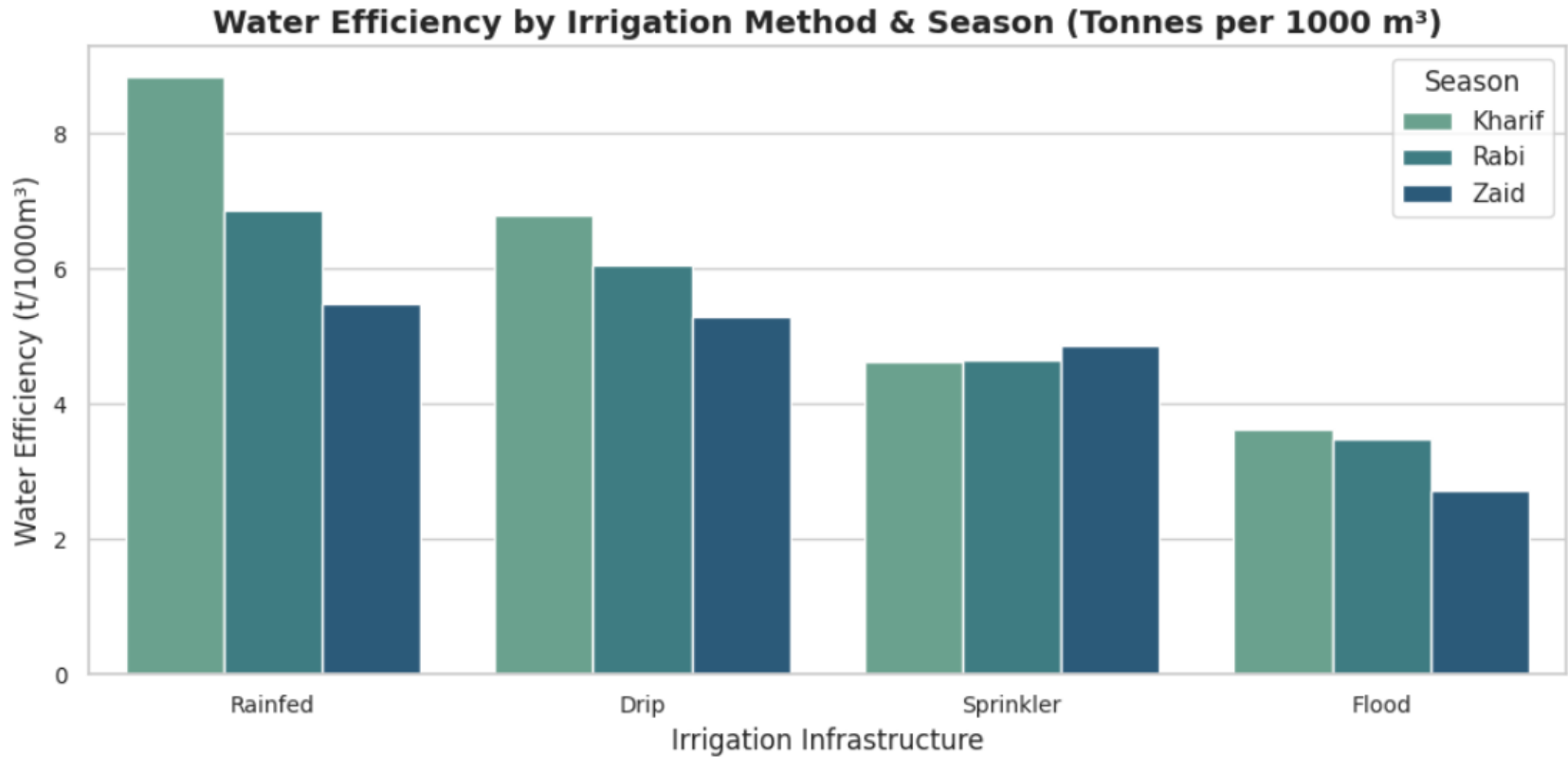


```
[7] ✓ 0s # Cell 7: Irrigation Performance & Water Efficiency
plt.figure(figsize=(10, 5))
order = (
    df_clean.groupby("Irrigation_Method")["Water_Efficiency_t_per_1000m3"]
        .mean()
        .sort_values(ascending=False)
        .index
)

sns.barplot(
    data=df_clean,
    x="Irrigation_Method",
    y="Water_Efficiency_t_per_1000m3",
    hue="Season",
    order=order,
    palette="crest",
    errorbar=None,
)

plt.title(
    "Water Efficiency by Irrigation Method & Season (Tonnes per 1000 m³)"
)

plt.xlabel("Irrigation Infrastructure")
plt.ylabel("Water Efficiency (t/1000m³)")
plt.show()
```

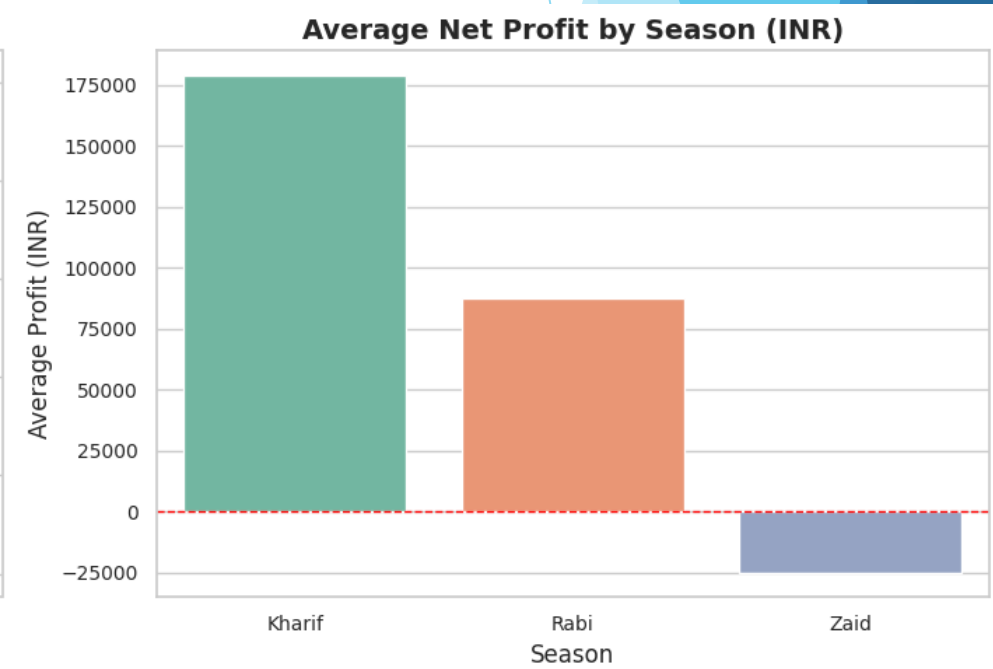
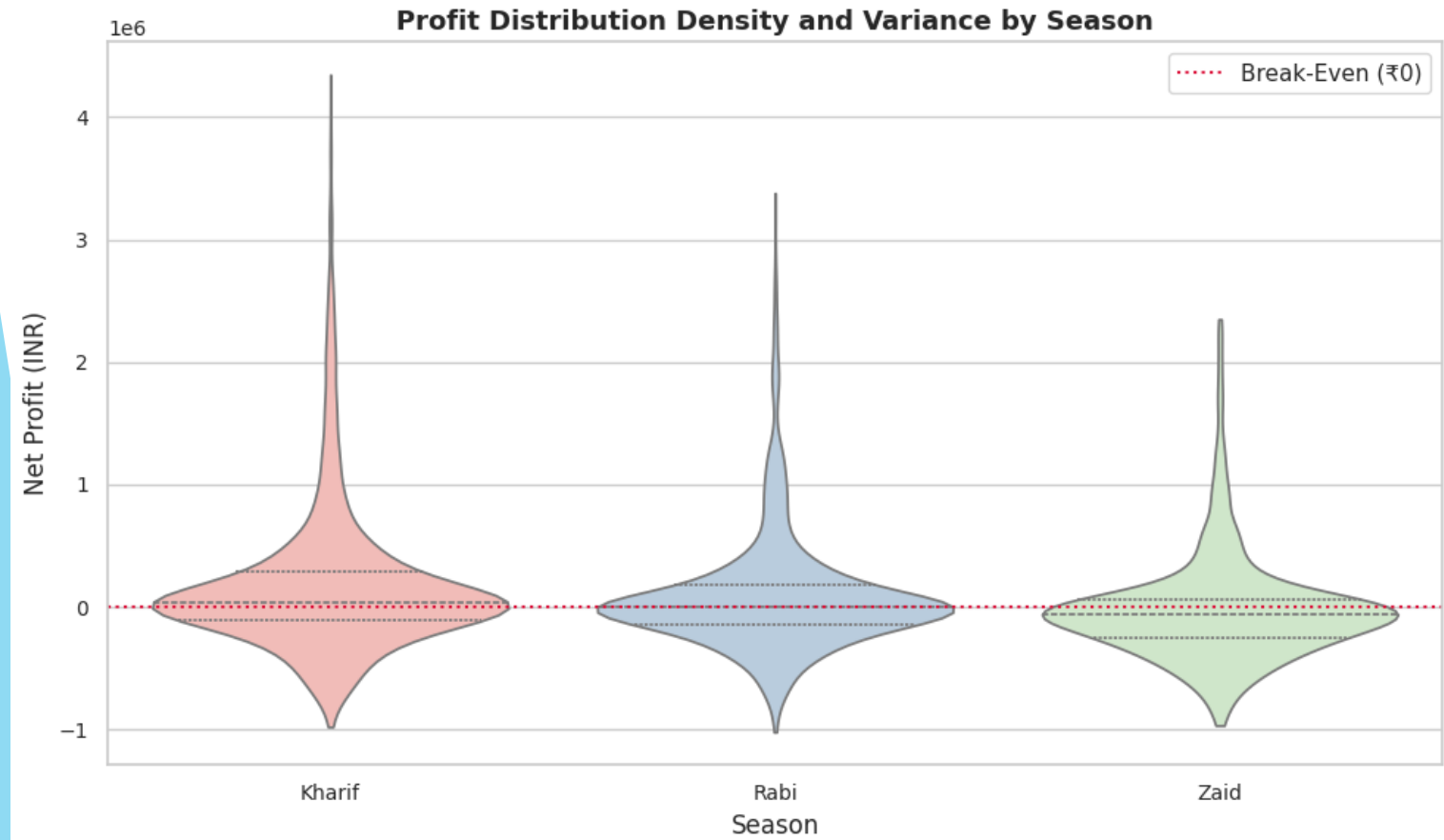


RESULTS

Economic & Net Profitability Disparities



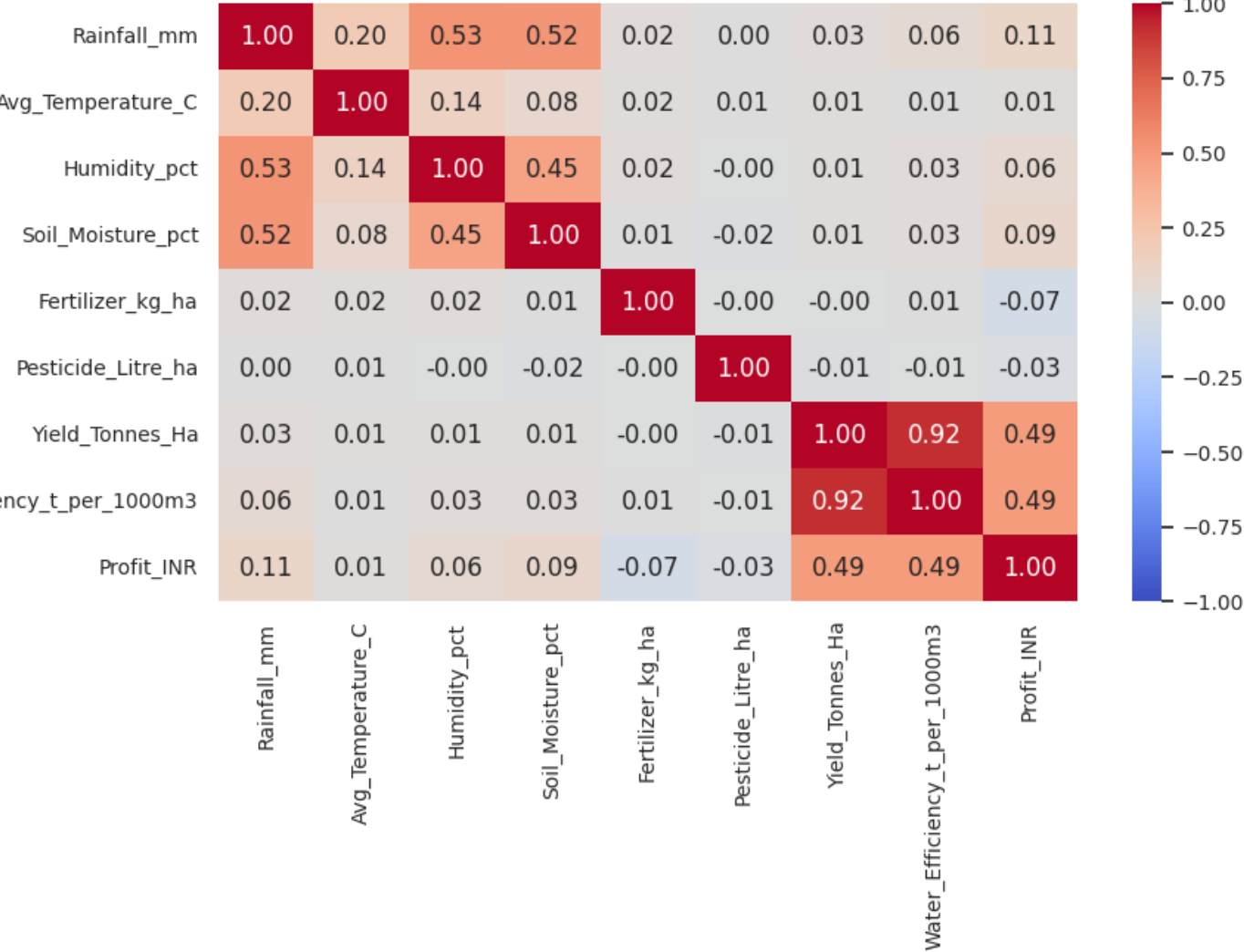
```
[10] ✓ 0s # Cell 10: Violin Plot of Net Profitability
plt.figure(figsize=(10, 6))
sns.violinplot(
    data=df_clean,
    x="Season",
    y="Profit_INR",
    palette="Pastel1",
    inner="quartile",
    cut=0,
)
plt.axhline(
    0, color="crimson", linestyle=":", linewidth=1.5, label="Break-Even (₹0)"
)
plt.title(
    "Profit Distribution Density and Variance by Season",
    fontsize=13,
    weight="bold",
)
plt.xlabel("Season")
plt.ylabel("Net Profit (INR)")
plt.legend(loc="upper right")
plt.show()
```



RESULTS

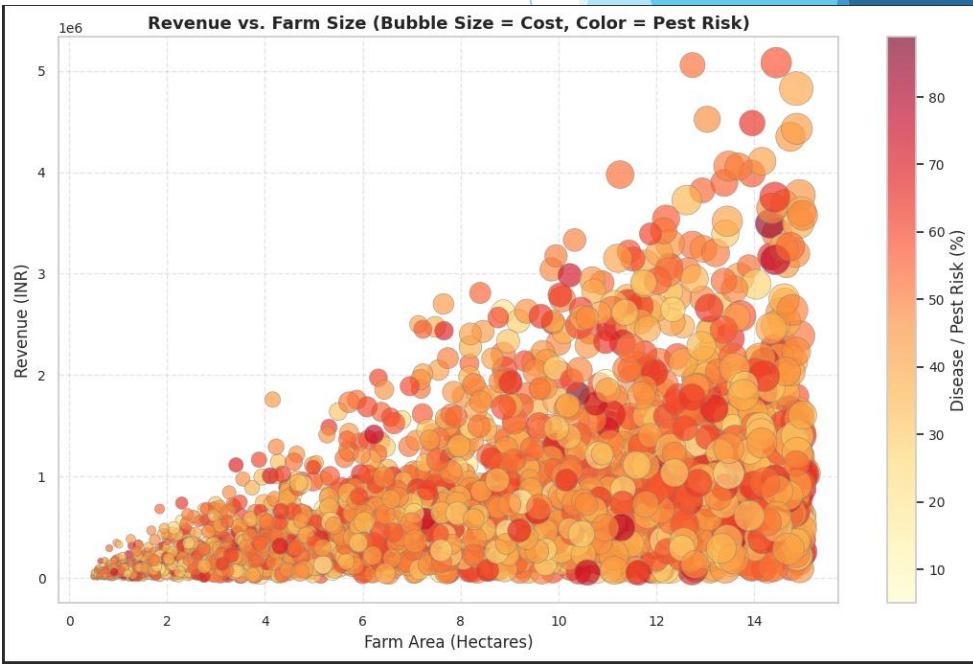
Agronomic Drivers & Code Implementation

Agronomic & Economic Factor Correlation Heatmap

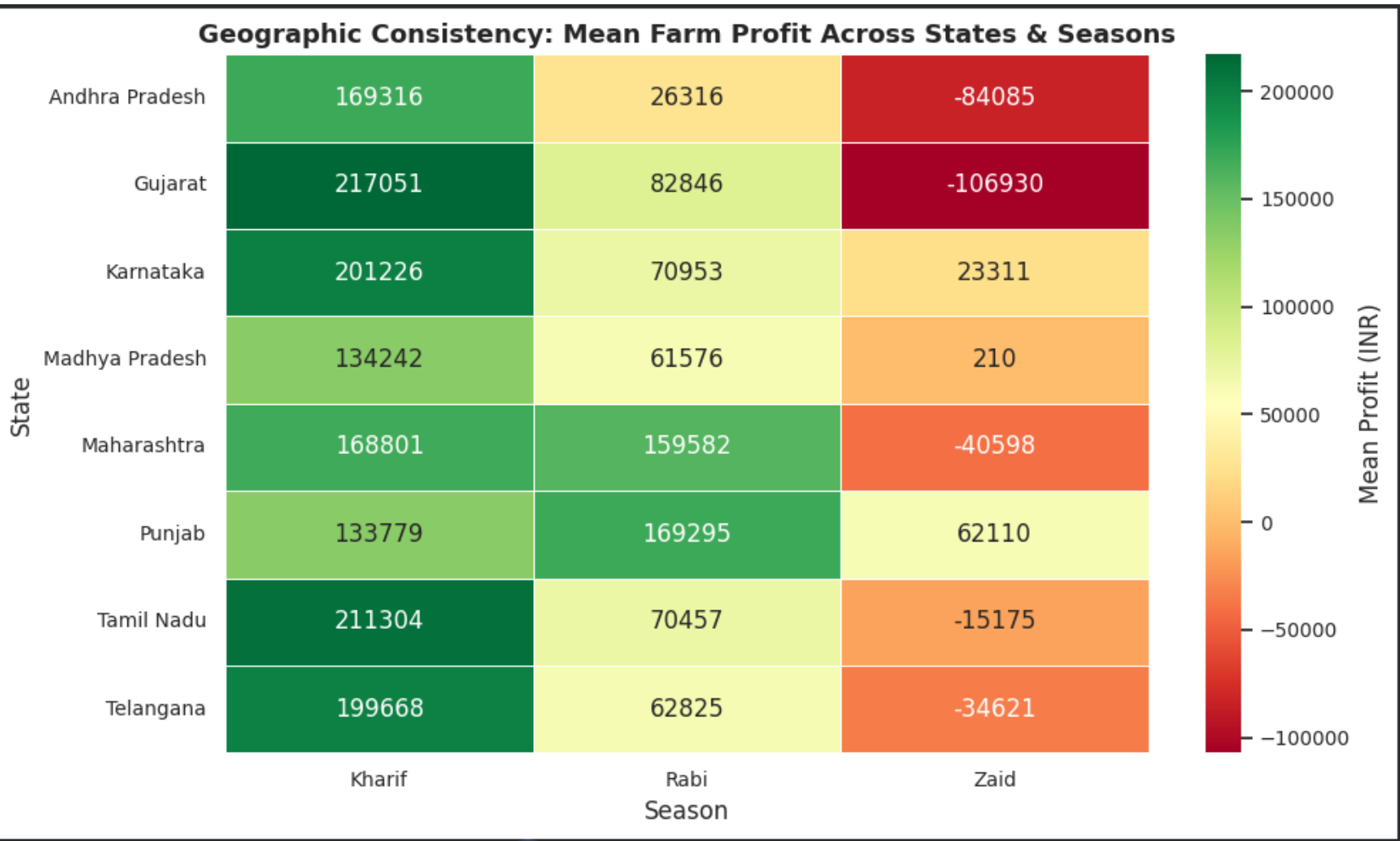


```
[8] ✓ 1s # Cell 8: Correlation Matrix
corr_cols = [
    "Rainfall_mm",
    "Avg_Temperature_C",
    "Humidity_pct",
    "Soil_Moisture_pct",
    "Fertilizer_kg_ha",
    "Pesticide_Litre_ha",
    "Yield_Tonnes_Ha",
    "Water_Efficiency_t_per_1000m3",
    "Profit_INR",
]

plt.figure(figsize=(10, 7))
corr_matrix = df_clean[corr_cols].corr()
sns.heatmap(
    corr_matrix, annot=True, cmap="coolwarm", fmt=".2f", vmin=-1, vmax=1
)
plt.title("Agronomic & Economic Factor Correlation Heatmap")
plt.show()
```



RESULTS



```
[ ] ▶ # Cell 14: Regional Consistency Analysis (State vs. Season Profitability)
state_season_pivot = df_clean.pivot_table(
    index="State", columns="Season", values="Profit_INR", aggfunc="mean"
).round(0)

plt.figure(figsize=(10, 6))
sns.heatmap(
    state_season_pivot,
    annot=True,
    fmt=".0f",
    cmap="RdYlGn",
    linewidths=0.5,
    cbar_kws={"label": "Mean Profit (INR)"},
)
plt.title(
    "Geographic Consistency: Mean Farm Profit Across States & Seasons",
    fontsize=13,
    weight="bold",
)
plt.xlabel("Season")
plt.ylabel("State")
plt.show()

display(state_season_pivot)
```

Season	Kharif	Rabi	Zaid
State			
Andhra Pradesh	169316.0	26316.0	-84085.0
Gujarat	217051.0	82846.0	-106930.0
Karnataka	201226.0	70953.0	23311.0
Madhya Pradesh	134242.0	61576.0	210.0
Maharashtra	168801.0	159582.0	-40598.0
Punjab	133779.0	169295.0	62110.0
Tamil Nadu	211304.0	70457.0	-15175.0
Telangana	199668.0	62825.0	-34621.0

Future scope

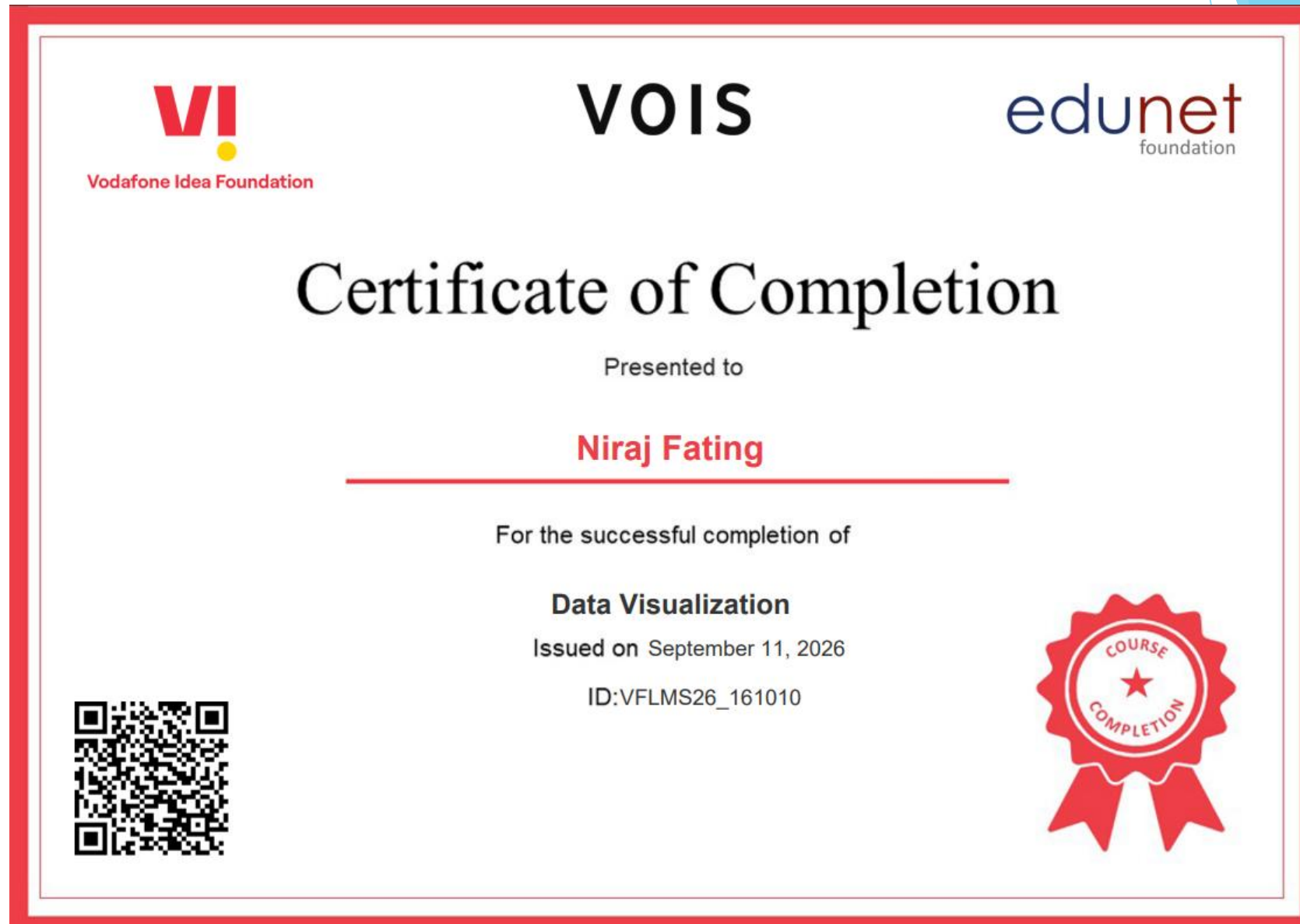
- **Predictive Machine Learning:** Train supervised regression models (XGBoost, Random Forest) and time-series models to forecast seasonal farm yield and crop market prices.
- **Real-Time IoT Integration:** Incorporate live sensor telemetry (measuring soil NPK levels, moisture, and ambient temperature) using lightweight MQTT protocols.
- **Automated Irrigation Decision Support:** Implement reinforcement learning models to trigger automated drip valve actuators based on real-time soil moisture thresholds.
- **Mobile Farmer Advisory Platform:** Deploy a vernacular Android / WhatsApp advisory bot delivering weekly crop health recommendations tailored to local weather forecasts.

GitHub Link

<https://github.com/Niraj-Fating/Seasonal-Agriculture-Performance-Analysis/tree/main>



**VOIS Course completion certificate -
Course name- Data Visualization**



Thank you